Experiment 1(a) - Analyze the trend of data science job postings

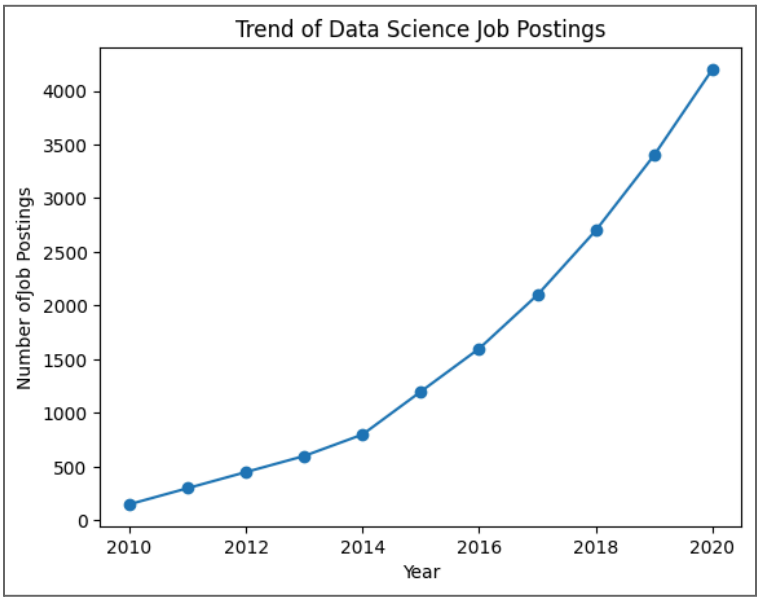
Aim:

To analyze the trend of data science job postings over the last decade.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number ofJob Postings')
plt.show()
```

Output:



Result:

Thus, the trend of data science job postings over the last decade has been analyzed.

Experiment 1(b) - Analyze and visualize the distribution of various data science roles

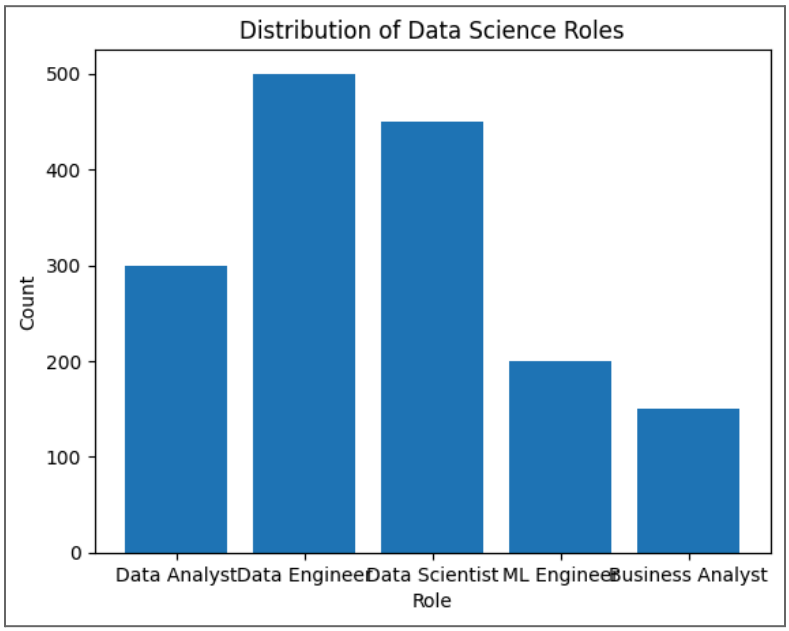
Aim:

To analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Output:



Result:

Thus, we have analyzed and visualized the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Experiment 1(c) - Conduct an experiment to differentiate structured and semi-structured data

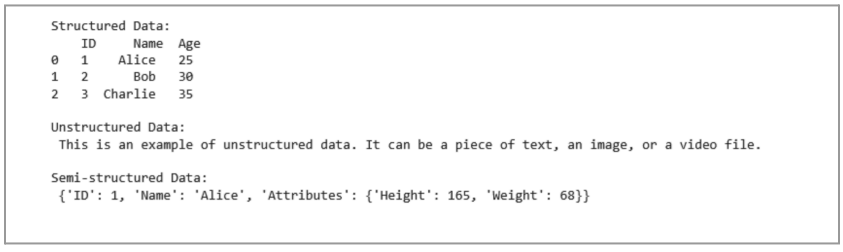
Aim:

To conduct an experiment to differentiate structured, unstructured and semi-structured data based on the data sets given.

Program:

```
import pandas as pd
# Structured data example
structured_data = pd.DataFrame (
   {"ID": [1, 2, 3], "Name": ["Alice", "Bob", "Charlie"], "Age": [25, 30, 35]} )
print("Structured Data:\n", structured_data)
# Unstructured data example
unstructured_data = """This is an example of unstructured data. It can be a piece of text, an image,
or a video file."""
print("\nUnstructured Data:\n", unstructured_data)
# Semi-structured data example (JSON)
semi_structured_data = {
   "ID": 1,
   "Name": "Alice",
   "Attributes": {"Height": 165, "Weight": 68},
}
print("\nSemi-structured Data:\n", semi_structured_data)
```

Output:



```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

Result:

Thus, we have conducted an experiment to differentiate between structured, unstructured, and semi-structured data based on the given data sets.

Experiment 1(d) - Conduct an experiment to encrypt and decrypt given sensitive data. Use the cryptography library fernet.

Aim:

To conduct an experiment to encrypt and decrypt given sensitive data.

Program:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b 'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

Output:

```
Original Data: b'Hello World!'
Encrypted Data: b'gAAAAABqhIP3GL7wWfQXx8L_FRQHiy_RHxhpP9Uk4TISCkHBjGdva3SKgBzC1eDBXsfaGadD3EDSJceEf_z52kzrarwcB2TgIg=='
Decrypted Data: b'Hello World!'
```

Result:

Thus, we have conducted an experiment to encrypt and decrypt the given sensitive data.